

Independence

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Definition 0.1 (Statistical independence). Random variables $X_1, \dots, X_n$ are **statistically independent** if, for all sets of real numbers $A_1, \dots, A_n$, the [probability](#) that every X_i falls in its set is the product of the individual probabilities:

$$\Pr(X_1 \in A_1, \dots, X_n \in A_n) = \prod_{i=1}^n \Pr(X_i \in A_i)$$

We write $X \perp\!\!\!\perp Y$ for “ X and Y are independent”. The symbol $\perp\!\!\!\perp$ is essentially $\prod$ upside-down, which can remind you of the definition.

For discrete random variables, independence is equivalent to the [joint PMF](#) factoring, $\Pr(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \Pr(X_i = x_i)$ for all $x_1, \dots, x_n$. For continuous random variables with a [joint density](#), it is equivalent to the joint density factoring the same way, and likewise for a [joint density-mass function](#) when one variable is discrete and the other continuous. The PMF form does not work as a definition for continuous random variables: there, both sides are 0 at every point, so it would call every pair of continuous random variables independent.

Example 0.1 (Two fair coin flips). Flip two fair coins, and let X_1 and X_2 indicate heads on the first and second flip. Each of the four outcomes has probability $1/4$, so for example:

$$\Pr(X_1 = 1, X_2 = 1) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \Pr(X_1 = 1) \Pr(X_2 = 1)$$

and the same factorization holds for the other three pairs of values, so $X_1 \perp\!\!\!\perp X_2$. In contrast, X_1 and the total number of heads, $X_1 + X_2$, are not independent: $\Pr(X_1 = 0, X_1 + X_2 = 2) = 0$, but $\Pr(X_1 = 0) \Pr(X_1 + X_2 = 2) = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$.

Definition 0.2 (Conditional independence). Random variables $Y_1, \dots, Y_n$ are **conditionally independent** given a discrete random variable (or vector) $\tilde{X}$ if, for every value $\tilde{x}$ with $\Pr(\tilde{X} = \tilde{x}) > 0$ and all sets of real numbers $A_1, \dots, A_n$, the conditional probability that every Y_i falls in its set is the product of the individual conditional probabilities:

$$\Pr(Y_1 \in A_1, \dots, Y_n \in A_n \mid \tilde{X} = \tilde{x}) = \prod_{i=1}^n \Pr(Y_i \in A_i \mid \tilde{X} = \tilde{x})$$

When $\tilde{X}$ is continuous, every event $\{\tilde{X} = \tilde{x}\}$ has probability 0, so the condition is stated with densities instead: for every $\tilde{x}$ with $p(\tilde{X} = \tilde{x}) > 0$ and all $y_1, \dots, y_n$,

$$\frac{p(\tilde{X} = \tilde{x}, Y_1 = y_1, \dots, Y_n = y_n)}{p(\tilde{X} = \tilde{x})} = \prod_{i=1}^n \frac{p(\tilde{X} = \tilde{x}, Y_i = y_i)}{p(\tilde{X} = \tilde{x})}$$

where each $p(\cdot)$ is a [joint density](#) (with probability masses in place of densities for any discrete Y_i , as in a [joint density-mass function](#)).

In regression, $\tilde{X}$ is usually the full set of covariates $(X_1, \dots, X_n)$, and a model usually also assumes that each Y_i depends on the covariates only through its own X_i : $\Pr(Y_i \in A_i \mid X_1 = x_1, \dots, X_n = x_n) = \Pr(Y_i \in A_i \mid X_i = x_i)$. Together, the two assumptions give the factorization $\Pr(Y_1 \in A_1, \dots, Y_n \in A_n \mid X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \Pr(Y_i \in A_i \mid X_i = x_i)$. That second assumption is a separate one, not part of conditional independence.

For continuous $\tilde{X}$, each ratio in the density form is the density of the Y 's under their conditional distribution given $\tilde{X} = \tilde{x}$, so both forms say the same thing: given $\tilde{X} = \tilde{x}$, the Y_i 's joint distribution factors.

Conditional independence neither implies nor is implied by independence.

Example 0.2 (Two tests of the same patient). Let X indicate whether a patient has a disease, and let Y_1 and Y_2 indicate positive results on two tests whose errors are unrelated, so that Y_1 and Y_2 are conditionally independent given X . Suppose $\Pr(X = 1) = 0.1$, each test is positive with probability 0.9 if $X = 1$ and with probability 0.1 if $X = 0$. Then, by the [law of total probability](#):

$$\begin{aligned} \Pr(Y_1 = 1, Y_2 = 1) &= (0.9)(0.9)(0.1) + (0.1)(0.1)(0.9) && \text{(condition on } X; \text{ factor given } X) \\ &= 0.081 + 0.009 && \text{(multiply)} \\ &= 0.09 && \text{(add)} \end{aligned}$$

but $\Pr(Y_1 = 1) = (0.9)(0.1) + (0.1)(0.9) = 0.18$, so $\Pr(Y_1 = 1) \Pr(Y_2 = 1) = 0.0324 \neq 0.09$: the tests are conditionally independent given X , but not independent.

Example 0.3 (Two coins, given their total). Let Y_1 and Y_2 indicate heads on two fair coin flips, which are independent ([Example 0.1](#)), and let $X = Y_1 + Y_2$ be the number of heads. Given $X = 1$, which has probability $1/2$, exactly one coin is heads, so:

$$\begin{aligned}
\Pr(Y_1 = 1, Y_2 = 1 \mid X = 1) &= \frac{\Pr(Y_1 = 1, Y_2 = 1, X = 1)}{\Pr(X = 1)} && \text{(definition of conditional probability)} \\
&= \frac{0}{1/2} && \text{(two heads make } X = 2, \text{ not } 1) \\
&= 0 && \text{(divide)}
\end{aligned}$$

but

$$\begin{aligned}
\Pr(Y_1 = 1 \mid X = 1) &= \frac{\Pr(Y_1 = 1, X = 1)}{\Pr(X = 1)} && \text{(definition of conditional probability)} \\
&= \frac{\Pr(Y_1 = 1, Y_2 = 0)}{\Pr(X = 1)} && (Y_1 = 1 \text{ and } X = 1 \text{ means } Y_2 = 0) \\
&= \frac{1/4}{1/2} && \text{(substitute)} \\
&= \frac{1}{2} && \text{(divide)}
\end{aligned}$$

and likewise $\Pr(Y_2 = 1 \mid X = 1) = 1/2$, so the product of the conditional probabilities is $1/4 \neq 0$: Y_1 and Y_2 are independent, but not conditionally independent given X .

Definition 0.3 (Identically distributed). Random variables $X_1, \dots, X_n$ are **identically distributed** if they all have the same [CDF](#):

$$\forall i \in \{1, \dots, n\}, \forall x \in \mathbb{R} : \Pr(X_i \leq x) = \Pr(X_1 \leq x)$$

For discrete random variables, this is equivalent to all of them having the same PMF. As with independence, the PMF form would not work as a definition in general: every continuous random variable has $\Pr(X_i = x) = 0$ at every x .

Example 0.4 (Identically distributed but not independent). In [Example 0.1](#), X_1 and $1 - X_1$ (the indicator of tails on the first flip) both take the values 0 and 1 with probability $1/2$ each, so they are identically distributed. They are not independent: knowing one determines the other.

Definition 0.4 (Conditionally identically distributed). Random variables $Y_1, \dots, Y_n$ are **conditionally identically distributed** given random variables $X_1, \dots, X_n$ if the conditional CDF of each Y_i given $X_i = x$ is one shared function $G(y \mid x)$ of y and x : for every $i \in \{1, \dots, n\}$, every y , and every x with $\Pr(X_i = x) > 0$,

$$\Pr(Y_i \leq y \mid X_i = x) = G(y \mid x)$$

When X_i is continuous, the condition applies at every x with $p(X_i = x) > 0$, with $\Pr(Y_i \leq y \mid X_i = x)$ computed from densities: the integral of $p(X_i = x, Y_i = u)/p(X_i = x)$ over $u \leq y$ (a sum over values $u \leq y$ when Y_i is discrete).

Example 0.5 (A shared regression model). Suppose each Y_i is binary, with $\Pr(Y_i = 1 \mid X_i = x) = \pi(x)$ for one function π shared by every i (for instance, $\pi(x) = x/(1+x)$ for a dose $x \geq 0$). Then $Y_1, \dots, Y_n$ are conditionally identically distributed given $X_1, \dots, X_n$, with $G(y \mid x) = 1 - \pi(x)$ for $0 \leq y < 1$ (and 0 for $y < 0$, 1 for $y \geq 1$). Their marginal distributions can still differ: if participant 1 always receives dose 0 and participant 2 always receives dose 1, then $\Pr(Y_1 = 1) = 0$ but $\Pr(Y_2 = 1) = 1/2$.

Definition 0.5 (Independent and identically distributed). Random variables $X_1, \dots, X_n$ are **independent and identically distributed** (shorthand: “ X_i iid”) if they are:

- statistically independent, and
- identically distributed.

The IID assumption is one of the most common assumptions in introductory statistics: it says a sample $X_1, \dots, X_n$ can be treated as n independent draws from a single shared distribution.

Example 0.6 (Repeated die rolls). The results of n rolls of the same fair die are IID: the rolls are independent, and each is uniform on $\{1, \dots, 6\}$.

Definition 0.6 (Conditionally independent and identically distributed). Random variables $Y_1, \dots, Y_n$ are **conditionally independent and identically distributed** given $X_1, \dots, X_n$ (shorthand: “ $Y_i \mid X_i$ ciid” or just “ $Y_i \mid X_i$ iid”) if:

- $Y_1, \dots, Y_n$ are conditionally independent given $(X_1, \dots, X_n)$,
- each Y_i depends on $(X_1, \dots, X_n)$ only through X_i , and
- $Y_1, \dots, Y_n$ are conditionally identically distributed given $X_1, \dots, X_n$.

Example 0.7 (The usual regression assumption). In Example 0.5, if the Y_i are also conditionally independent given all the doses, and each Y_i depends on the doses only through its own X_i , then $Y_i \mid X_i$ ciid, and the joint conditional PMF is $\prod_{i=1}^n \pi(x_i)^{y_i} (1 - \pi(x_i))^{1-y_i}$: one shared function evaluated at each (x_i, y_i) .